Categorical composable cryptography

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Composability in cryptography

One would expect that if you wire together "provably secure" protocols you end up with a secure protocol.

- This is false in general! Standard game-based security notions don't necessarily guarantee composability. In fact, many "secure" protocols might not be secure anymore if several copies are run concurrently.

- QKD and 20(ish) years between first security proofs and composable ones.

- Several frameworks for composability and plenty of work within them, but none have convinced the whole community.

Real-world ideal-world paradigm

AKA simulation paradigm.

Usual definition: a real protocol P securely realizes the ideal functionality F from the resource R if for any attack A on $P \circ R$ there is a simulator S on F such that $(A, P) \circ R$ is indistuingishable from $S \circ F$ by any (efficient) environment.

"Any bad thing that could happen during the protocol could also happen in the ideal world."

Usual ways of making this precise:

- Fixing a concrete low-level formalism for interactive computation (e.g. UC-security)
- Abstract cryptography and constructive cryptography — more on this next

Comparison with abstract & constructive cryptography

Hugely indebted to these in spirit even if not technically.
Some similarities:

- The simulation paradigm
- Cryptography as resource theory: the resources are various functionalities (e.g. keys, channels etc) and transformations are given by protocols that build the target resource *securely* from the starting resources. For instance, OTP is a protocol $\textit{key} \otimes \textit{insecure channel} \to \textit{secure channel}$
- Agnostic about the underlying computational model.
- The intuition of resources as "boxes with interfaces" available to various parties.
- Starting from perfect security, then relaxing to computational security either via an equivalence relation ("indistinguishability") or pseudometric ("distinguisher advantage")

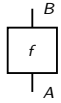
Comparison with abstract & constructive cryptography

The main difference is that we use use (symmetric monoidal or compact closed) categories as our algebraic theory of systems. I believe this approach comes with some benefits:

- Monoidal cats have sequential and parallel composition as their primitives — just like cryptographers want to compose their protocols
- Attack models as an explicit part of the formalism—perhaps we can reason composably about multiple kinds of attackers?
- Graphical calculus as an intuitive yet rigorous tool—e.g. pictorial proofs of (known) no-go results
- Intial attacks vs Canetti's "completeness of the dummy adversary"
- Abstract version of Unruh's lifting theorems.
- Other (potential) connections to TCS and math

Categories via pictures

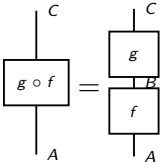
A category has objects and morphisms between them. We will depict a morphism $f : A \to B$ by



Special morphisms get special pictures. For plain categories there's only the identity morphisms $\mathrm{id}_A : A \to A$ drawn as

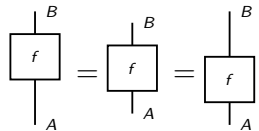


Morphisms can be composed if the object in the middle matches:



Categories via pictures

The pictures make the axioms implicit, or alternatively - the axioms are whatever is needed for the pictures to be unambiguous. Besides associativity, we have left and right unit laws:

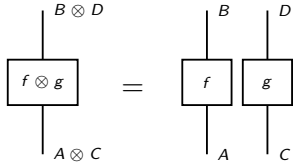


Example

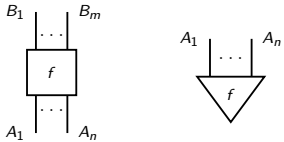
- Sets-with structure and structure-preserving-maps: e.g. sets and functions, topological spaces and cont. maps, vector spaces and linear maps
- Any group gives a category with a single object: morphisms correspond to elements of G and composition is given by multiplication in G
- Any poset $(X, \leq)$ gives a category: elements of X are the objects, and there's a single morphism $x \to y$ iff $x \leq y$

Monoidal categories via pictures

In a *monoidal* category one also has parallel composition of objects and morphism



The monoidal unit is denoted by the empty picture, and boxes can now have arbitrary amounts of input and output wires (a box with no inputs is called a state):



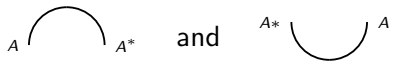
Examples: $(\mathbf{Set}, \times, 1)$, $(\mathbf{Vect}_{\mathbb{F}}, \otimes, \mathbb{F})$

Symmetry and compact closure via pictures

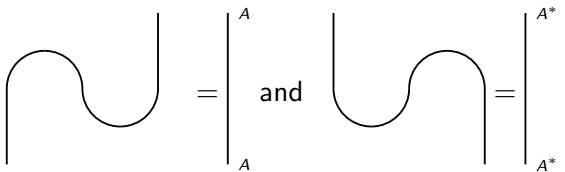
In a *symmetric* monoidal category (SMC) one can also cross wires, and only connectivity matters.



In a compact closed category we also have cups and caps, i.e. morphisms

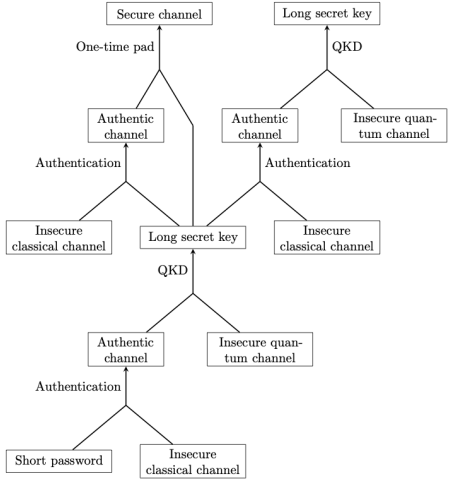


satisfying the snake equations



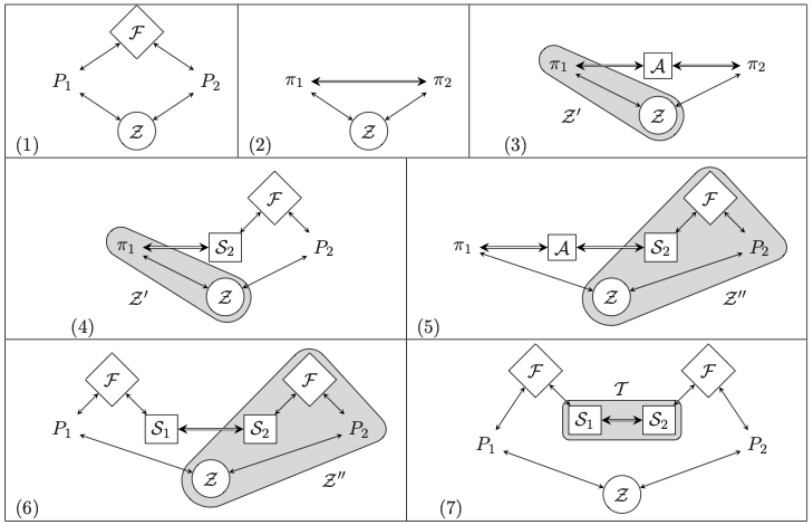
This blurs the distinction between inputs and outputs. Eg. $(\mathbf{Rel}, \times, 1)$, $(\mathbf{FVect}_{\mathbb{F}}, \otimes, \mathbb{F})$

CT in cryptography I: protocols as morphisms



Source: Portmann & Renner (2014) - Cryptographic security of quantum key distribution

CT in cryptography II: reasoning about protocols



Source: Rosulek & Prabhakaran (2008) - Cryptographic Complexity of Multi-Party Computation Problems

Resource theories

Examples:

- Can these noisy channels be used to simulate a (almost) noiseless channel?
- Is there a LOCC-protocol that transforms this quantum state to that one?
- Can these (possibly non-local) correlations be used to simulate those?
- Any preordered commutative monoid.

For a category theorist, this is roughly speaking an SMC where you mostly care whether a transformation $A \to B$ exists or not.

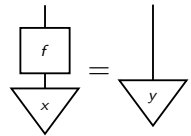
Resource theory of states

In '*A mathematical theory of resources*'

Coecke, Fritz & Spekkens, Information and Computation (2016).

many resource theories are built theories starting from an SMC $\mathbf{C}$ equipped with a (wide) sub-SMC $\mathbf{C}_F$ of free processes.

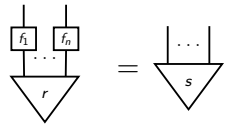
One of the constructions – the resource theory of states – can be desribed as follows: the resources are states of $\mathbf{C}$ and the resource conversions $x \to y$ are maps f in $\mathbf{C}_F$ such that



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Resource theory of n-partite states

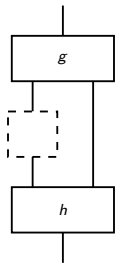
n-partite version: Objects are of the form $((A_i)_{i=1}^n, r\colon I \to \bigotimes A_i)$. A map $((A_i)_{i=1}^n, r) \to ((B_i)_{i=1}^n, s)$ is then a tuple $(f_i)_{i=1}^n$ that transforms r to s:



We think of this as a resource theory with n parties who try to agree on actions $f_1, \dots f_n$ to transform some resource to another one.

Resource theory of maps

We can easily vary the construction to have our resources be arbitary morphisms $f : A \to B$ (or $f : \bigotimes_{i=1}^{n} A_i \to \bigotimes_{i=1}^{n} B_i$ in the n-partite case) and our resource conversions be given by (n-tuples of) "combs"



built out of free processes.

In general, it suffices to have a "well-behaved"[1] operation R assigining to each object A the set $R(A)$ of resources of type A and to each $f : A \to B$ a function $R(f) : R(A) \to R(B)$ explaining how f transforms resources.

[1] lax monoidal functor

Towards security

Such protocols have no security guarantees—what if some subset of the parties does something else instead?

If a subset J of $[n] := \{1, \dots n\}$ is malicious, they can replace f_js for $j \in J$ with anything. The simulation paradigm says that the protocol is secure $r \to s$ if for any such attack on $(f_1 \dots f_n)$ the subset could've attacked s with the same end-result

We abstract from here:

- an abstract attack model $\mathcal{A}$ that gives for each protocol f a collection $\mathcal{A}(f)$ of attacks on it
- security against $\mathcal{A}$: for each attack on the protocol there is an attack on the target with similar end-results

Abstract attacks

Definition

An *attack model* $\mathcal{A}$ on an SMC $\mathbf{C}$ consists of giving for each morphism f of $\mathbf{C}$ a class $\mathcal{A}(f)$ of morphisms of $\mathbf{C}$ such that

1. $f \in \mathcal{A}(f)$ for every f.
2. For any $f\colon A \to B$ and $g\colon B \to C$ and composable $g' \in \mathcal{A}(g), f' \in \mathcal{A}(f)$ we have $g' \circ f' \in \mathcal{A}(g \circ f)$. Moreover, any $h \in \mathcal{A}(g \circ f)$ factorizes as $g' \circ f'$ with $g' \in \mathcal{A}(g)$ and $f' \in \mathcal{A}(f)$.
3. For any $f\colon A \to B$, $g\colon C \to D$ in $\mathbf{C}$ and $f' \in \mathcal{A}(f), g' \in \mathcal{A}(g)$ we have $f' \otimes g' \in \mathcal{A}(f \otimes g)$. Moreover, any $h \in \mathcal{A}(f \otimes g)$ factorizes as $h' \circ (f' \otimes g')$ with $f' \in \mathcal{A}(f)$, $g' \in \mathcal{A}(g)$ and $h' \in \mathcal{A}(\mathrm{id}_{B \otimes D})$.

Examples

- $\mathcal{A}_{\mathsf{min}}(f) := \{f\}$ — represents honest behavior
- $\mathcal{A}_{\mathsf{max}}(f) := \mathit{Mor}(\mathbf{C})$ — represents arbitrary malicious behavior
- If $\mathcal{A}_i$ is an attack model on $\mathbf{C}_i$, then $\prod \mathcal{A}_i$ is an attack model on $\prod_i \mathbf{C}_i$. For instance, $\mathcal{A}_{\mathsf{min}} \times \mathcal{A}_{\mathsf{max}}$ represents two parties, Alice and Bob, with Alice honest and Bob malicious.
- In a concrete model of probabilistic interacting computation, can set $\mathcal{A}(f) := \{$ honest-but-curious variants of $f\}$
- Very conjenctural: quantum analogues of HBC

Abstract security

Definition

Given $R\colon \mathbf{C} \to \mathbf{Set}$ and resources $r \in R(A)$, $s \in R(B)$, we say that $f\colon A \to B$ is a resource conversion $r \to s$, if $R(f)r = s$.
We say that a resource conversion f is *secure* against an attack model $\mathcal{A}$ on $\mathbf{C}$ (or $\mathcal{A}$-secure) if for any $f' \in \mathcal{A}((f)$ with $\mathrm{dom}(f') = A$ there is $b \in \mathcal{A}(\mathrm{id}_B)$ such that $R(f')r = R(b)s$.

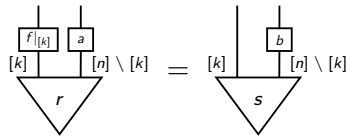
A subset X of $\mathcal{A}(f)$ is said to be *initial* if any $f' \in \mathcal{A}(f)$ with $\mathrm{dom}(f') = A$ can be factorized as $b \circ a$ with $a \in X$ and $b \in \mathcal{A}(\mathrm{id}_B)$.

Proposition

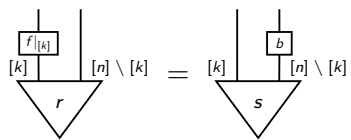
It suffices to check security against initial sets of attacks.

Examples

Assume the first k parties are honest and the last $n-k$ parties are dishonest. Then $(f_1, \dots f_k)$ is secure if for any a there is a b such that



It suffices to check this for the initial attack $\bigotimes_{k+1}^{n} \mathrm{id}$:



Initial honest-but-curious: follows the protocol and retains a transcript of it. Security: an identical (indistinguishable) protocol transcript can be simulated from the target functionality.

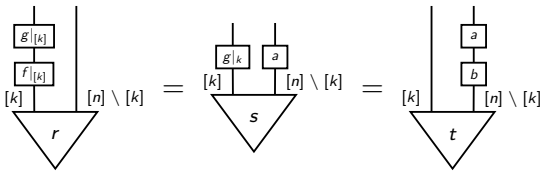
Composability

Theorem

Secure protocols form an SMC

Proof.

$\otimes$ and $\circ$ inherited form the ambient category—one just needs to check that they work. Here's the key steps for $\circ$ and $\otimes$ in the n-partite case with the first k parties honest



□

Security against multiple attack models

Corollary

Protocols secure against $\mathcal{A}_1, \ldots \mathcal{A}_k$ form a symmetric monoidal category

Proof.

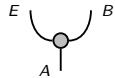
Symmetric monoidal subcategories are closed under intersection □

Example

Fix a family of subsets of $[n]$ parties: protocols secure against each of these subsets behaving maliciously form an SMC. For instance, in MPC one often studies protocols secure against at most $n/2$ or $n/3$ malicious participants.

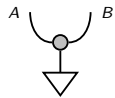
OTP: starting resources

Channel from Alice to Bob that leaks everything to Eve:



(Note: if instead the message goes via Eve (who may tamper with it), the analysis is different)

Shared random key:



Target resource: a channel



Free building blocks: local (efficient) computation

Local ingredients for OTP

A group structure on the message space: a multiplication  with unit  satisfying the following equations.

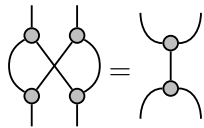


Note that copying and deleting satisfy similar equations

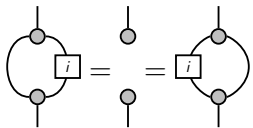


Rest of the group structure

In addition, multiplication and copying interact:

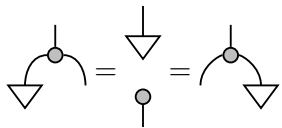


and the map  giving inverses satisfies



Uniform randomness

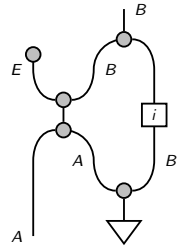
The key being uniformly random is captured by



"Adding uniform noise to a channel gives uniform noise"

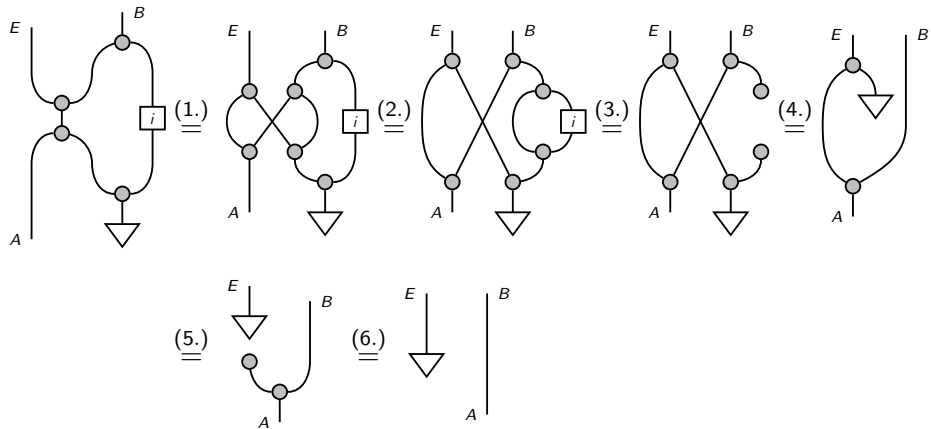
For the experts: a Hopf algebra with an integral in a symmetric monoidal category.

The protocol



Alice adds the key to her message, broadcasts it to Eve and Bob. Eve deletes her part and Bob adds the inverse of the key to recover the message.

Security of OTP

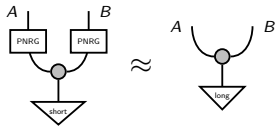


1. Bialgebra. 2. Associativity. 3. Antipode 4. Units 5. Random noise 6. Units.

More on OTP

In other words, anything Eve might learn from the ciphertext she could already compute without it, so this protocol is indeed a secure transformation against Eve.

Reusing keys is not a secure map $key \to key \otimes key$. However, a computationally secure PRNG will give a computationally secure way of constructing a long shared key from a short one, depicted by



where $\approx$ stands for computational indistinguishability.
Composing these two results in *the stream cipher*, which is secure automatically as a composite of secure protocols inside our framework.

Extensions of the simple model

The above captures a very particular cryptographic situation:
There is no set-up, i.e., the parties have no free cryptographic primitives or communication not given by the starting functionality.

- This can be fixed by fixing a class $\mathcal{X}$ of free resources and defining general protocols $r \to s$ as those of the form $r \otimes x \to s$.

Security is perfect (i.e. information theoretic) instead of computational. This can be fixed in two ways:

- replace $=$ with an equivalence relation $\approx$ modelling computational indistinguishability
- Work with a pseudometric, and work with approximately or asymptotically secure protocols

A no-go theorem for two parties

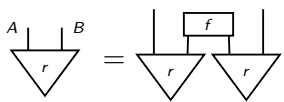
Let **C** now be a compact closed category, with $\cup$ modelling a shared communication channel.

Theorem

For Alice and Bob (one of whom might cheat), if a bipartite functionality r



can be realized from a communication channel between them, i.e. from $\cup$ *by a secure protocol, then r satisfies*

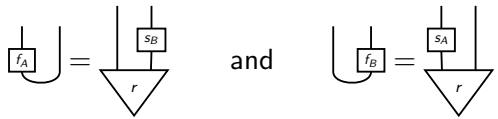


for some f .

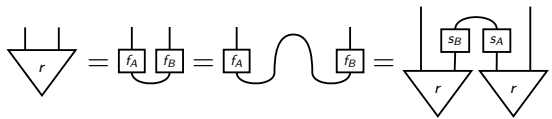
A no-go theorem for two parties

Proof.

Assume a protocol $f_A \otimes f_B$ achieving this. Security constraints against each party give us



Which gives

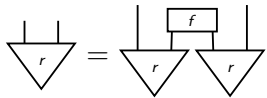


□

A no-go theorem for two parties

Theorem

For Alice and Bob (one of whom might cheat), if a bipartite functionality r can be realized from a communication channel between them, i.e. from $\cup$ by a secure protocol, then r satisfies



for some f.

Corollary

In the same bipartite setting, (composable) bit commitment and oblivious transfer are impossible without setup.

Summary

We have a categorical framework where

- composability is guaranteed (also for computational security)
- attack models are general enough to cover various kinds of adversarial behavior (e.g. colluding vs independent attackers)
- string diagrams can be used to make existing (or new) pictures into rigorous proofs

Questions...

?

Broadbent A., MK, "Categorical composable cryptography" (2021), arXiv:2105.05949

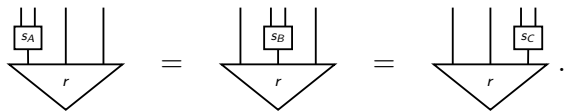
See also:

- my talk at Structure meets power workshop: slides, talk
- my talk at the Applied Category Theory conference: slides, talk

Bonus: tripartite no-go

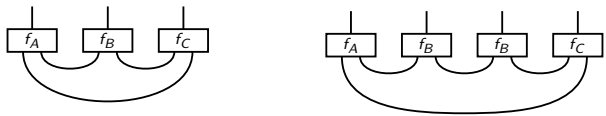
Theorem

If a tripartite functionality r can be realized from each pair of parties sharing a state$\cup$, securely against any single party, then there are simulators s_A, s_B, s_C such that



Bonus: tripartite no-go

Proof.
Any tripartite protocol building on top of each pair of parties sharing $\cup$ can be drawn as in the left side of

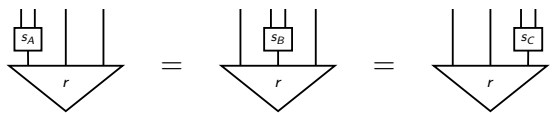


Consider now the picture depicted on the right. It can be seen as the result of three different attacks on the protocol: one where Alice cheats and performs f_A and f_B (and the wire connecting them), one where Bob performs f_B twice, and one where Charlie performs f_B and f_C. The security of (f_A, f_B, f_C) against each of these gives the required simulators. $\square$

Bonus: tripartite no-go

Theorem

If a tripartite functionality r can be realized from each pair of parties sharing a state$\cup$, securely against any single party, then there are simulators s_A, s_B, s_C such that



Corollary

Given a SMC **C** *modeling interactive computation, and a state$\cup$ on $A \otimes A$ modeling pairwise communication, it is impossible to build broadcasting channels securely (even asymptotically in terms of distinguisher advantage) from pairwise channels.*